University of Central Punjab
Faculty of IT &CS
Department of Applied Computing &Technologies
Subject: Agent Based Modeling

Due Date: 11-October 2026 **Assignment # 1** **Total Marks: 40**

Name of Student: **Registration No:** **Section:**

CLO#	Course Learning Outcome (CLO)	Taxonomy	PLO
1	**Explain** the concepts, architectures, evolution, reasoning, planning, and behavior of intelligent autonomous agents, LLMs, and multi-agent systems.	C2	P2

Instructions:

1. Please provide a complete solution. (Group Assignment)

2. There will be a viva for the assignment.

3. Any plagiarism or cheating, even in a single question, will result in zero marks for the entire assignment.

Title page of Assignment must include: *Course Name, Assignment no., Your Full name, Complete Roll no., Section, and Instructor's full name.*

Important Note: *Submit assignment timely from portal, through email assignment would not be acceptable.*

Overview

The purpose of this assignment is to understand how different AI agent architectures solve the same problem in different ways. You will design and implement seven single-agent types for one common food-delivery scenario and compare their behaviour. Multi-Agent Systems (MAS) are not required.

Scenario: Smart Food Delivery Agent

A food-delivery company wants an intelligent agent to help manage a delivery. The agent receives information about the order, delivery distance, traffic, weather, delivery time, available routes, and previous delivery experience.

Example: A customer places an order. Route A is shorter but has heavy traffic, while Route B is longer but has light traffic. It is raining. The agent must decide how to complete the delivery efficiently.

Task

Implement all six agent types below. Each implementation should clearly demonstrate the decision-making behaviour associated with that agent type.

Agent Type	Required Behaviour
1. Simple Reflex Agent	Use IF-THEN rules. Example: IF traffic is heavy, select an alternative route.
2. Model-Based Reflex Agent	Maintain an internal state such as current route, traffic condition, weather, and order status before deciding.

3. Goal-Based Agent	Choose actions that help achieve the goal: deliver the order successfully within the required time.
4. Utility-Based Agent	Compare possible routes using factors such as delivery time, distance, traffic, and cost, then select the option with the best utility.
5. Learning Agent	Use previous delivery results or feedback to improve future route or delivery decisions.
6. Hierarchical Agent	Break the main delivery goal into smaller tasks, such as accept order, choose route, pick up food, deliver order, and confirm delivery.
7. Contextual Agent	Collect and use contextual informaiotn.

Architecture Diagram

Create one simple architecture diagram for each agent type. The diagram should show how information moves through the agent and how the final action is selected.

Basic structure: Input / Perception → Decision Process → Action → Environment

Add the relevant component for each architecture, such as Internal State, Goal, Utility, Learning/Feedback, or Subgoals.

Testing

Test all six agents using the same three delivery situations:

1. Normal traffic and good weather.
2. Heavy traffic with multiple available routes.
3. Heavy traffic, rain, and an urgent delivery.

Deliverables

- Implementation/code for all six agent types.
- One architecture diagram for each agent type.
- Outputs/results for the three common test cases.
- A brief comparison explaining how the six agents behaved differently.
- A short conclusion identifying which agent type was most suitable for the scenario and why.

Main Objective

To practically understand how changing an AI agent's architecture changes the way it uses information, makes decisions, and selects actions in the same environment.